

Module 1: Vectors and Matrix Operations

1 Linear Algebra Review

The purpose of this section is to give a fly-by review of the basic facts about Linear Algebra that are needed for this course. You will not review absolutely everything at the front end so that we can get to some fun stuff sooner

Definition. *Basics of Vectors.* $\mathbb{R}^n$ is a vector space, which is a collection of vectors along with a notion of vector addition and scalar multiplication. Addition of two vectors is done coordinate wise:

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

and scalar multiplication is applied to all coordinates, i.e., if $\alpha \in \mathbb{R}$ and $x \in \mathbb{R}^n$, then

$$\alpha x = \begin{bmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{bmatrix}$$

Definition. *Given a collection of vectors $\{x_1, \dots, x_k\} \subset \mathbb{R}^n$, a linear combination of these vectors is any vector of the form*

$$\alpha_1 x_1 + \dots + \alpha_k x_k = \sum_{i=1}^k \alpha_i x_i, \quad \alpha_1, \dots, \alpha_k \in \mathbb{R}$$

Note any such linear combination must be in $\mathbb{R}^n$ by the vector space properties.

1.1 Definition of Linear Spaces

A linear space (or vector space) is a set of vectors along with two operations: vector addition and scalar multiplication. A vector space must satisfy the following axioms:

- Addition: If u and v are vectors in the space, then $u + v$ is also in the space.
- Scalar multiplication: If u is a vector in the space and c is a scalar, then $c \cdot u$ is also in the space.

Examples of linear spaces include $\mathbb{R}^n$, the space of all real-valued vectors with n components, and $\mathbb{C}^n$, the space of complex-valued vectors.

1.2 Subspaces

A subspace of a linear space is a subset that is also a linear space under the same addition and scalar multiplication. For a subset W of a vector space V to be a subspace, it must:

- Contain the zero vector.
- Be closed under addition and scalar multiplication.

For example, the set of all vectors in $\mathbb{R}^3$ that lie in a plane through the origin is a subspace of $\mathbb{R}^3$.

1.3 Linear Span

The span of a set of vectors $v_1, v_2, \dots, v_k$ in a vector space V is the set of all possible linear combinations of these vectors:

$$\text{span}\{v_1, v_2, \dots, v_k\} = \{c_1 v_1 + c_2 v_2 + \dots + c_k v_k \mid c_1, c_2, \dots, c_k \in \mathbb{R}\}$$

The span of a set of vectors forms a subspace of V .

2 Linear Dependence and Independence

- A set of vectors $v_1, v_2, \dots, v_k$ is linearly independent if the only solution to:

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k = \mathbf{0}$$

is $c_1 = c_2 = \dots = c_k = 0$.

- If non-zero coefficients $c_1, c_2, \dots, c_k$ exist that satisfy the above equation, the vectors are linearly dependent.

Linear independence is crucial because independent vectors provide a unique way to represent any vector in the space.

2.1 Basis and Dimension

A basis for a vector space is a set of linearly independent vectors that span the entire space. Every vector in the space can be written as a unique linear combination of the basis vectors. The number of vectors in a basis is the dimension of the vector space.

For example, the standard basis for $\mathbb{R}^2$ is $\{(1, 0), (0, 1)\}$, and the dimension of $\mathbb{R}^2$ is 2.

2.2 Basics of Matrices

We may denote real-valued matrices $A \in \mathbb{R}^{m \times n}$ in a variety of ways, for example via its scalar entries $A = [a_{i,j}]_{i=1}^m \substack{m \\ n \\ j=1}$ or by its columns $A = [A_{:1} \dots A_{:n}]$.

Note that many data sets can be naturally represented as a matrix, and thus are typically thought of as an array of numbers, but we will see that it is of great utility to consider the linear algebraic structure of such data matrices rather than simply considering them as a collection of scalars.

First, let's discuss matrix-vector multiplication. Given a matrix $A \in \mathbb{R}^{m \times n}$ and a vector $x \in \mathbb{R}^n$, the product Ax is a vector in $\mathbb{R}^m$ whose i -th coordinate is given by

$$(Ax)_i = \sum_{j=1}^n a_{ij}x_j = \langle A_{i,:}, x \rangle$$

That is, the i -th entry of Ax is the inner product of the i -th row of A with x . However, we can also write Ax as a linear combination of its columns!!

$$Ax = \sum_{j=1}^n x_j A_{:,j}$$

Recall here that $A_{:,j} \in \mathbb{R}^m$ is the j -th column of A .

So we can either evaluate Ax via inner products or linear combinations of columns of A . One should note that both methods of computation require mn multiplications, so neither is necessarily better from that viewpoint.

Now for matrix-matrix multiplication. We may do this in two ways that both have useful information embedded in them. First of all, we recall that for the matrix-matrix product AB to be well defined, B must have exactly as many rows as A has columns. So if $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, then $AB \in \mathbb{R}^{m \times p}$. If B has more or less than n rows, then the product is not defined.

Matrix Multiplication via Inner Products: The first way we can write the product is as inner products similar to what we did for matrix-vector products:

$$(AB)_{ij} = \langle A_{i,:}, B_{:,j} \rangle = \sum_{k=1}^n A_{ik} B_{kj}$$

Matrix Multiplication via Outer Products: Alternatively, we may express the matrix-matrix product by what are called outer products of two vectors, namely

$$AB = \sum_{k=1}^n A_{:,k} B_{k,:} \tag{1}$$

Note here that $A_{:,k}$ is a column vector with dimension $(m \times 1)$ and $B_{k,:}$ is a row vector with dimension $(1 \times p)$, hence their product has dimension $(m \times p)$ as we expected. To see that this is equivalent to the first expression, note that if we want the (i, j) -th entry of AB from the outer product form we need to

evaluate $\sum_{k=1}^p A_{ik} B_{kj}$ which is $(AB)_{ij}$.

Equation (1) gives us an interesting first glimpse at the fundamental notion of a matrix being represented as the sum of rank one matrices, which are the building blocks of matrix spaces in general. To wit, let us now discuss the notion of the rank of a matrix.

Example 1. Let $A \in \mathbb{R}^{m \times n}$.

(1) The column space of A is $\text{Col}(A) := \text{span}(A_{:1}, \dots, A_{:n})$ which is a subspace of $\mathbb{R}^m$.

(2) The row space of A is $\text{Row}(A) := \text{span}(A_{1:}, \dots, A_{m:})$ which is a subspace of $\mathbb{R}^n$.

A fundamental fact of Linear Algebra is the following.

Theorem 4.3. For every $A \in \mathbb{R}^{m \times n}$,

$$\dim(\text{Col}(A)) = \dim(\text{Row}(A))$$

This leads to a well-defined notion of the rank of a matrix:

Example 2. Given $A \in \mathbb{R}^{m \times n}$, the rank of A is given by

$$\text{rank}(A) := \dim(\text{Col}(A))$$

or equivalently (on account of Theorem 1.15) by $\text{rank}(A) = \dim(\text{Row}(A))$.

One observation is immediate since the dimensions of the row and column space cannot be more than m or n , respectively:

$$\text{rank}(A) \leq \min\{m, n\}$$

We also have the useful facts that

$$\text{Col}(A) = \text{Row}(A^\top), \quad \text{and} \quad \text{Row}(A) = \text{Col}(A^\top)$$

2.3 Content Curated Video

- **Linear Algebra:** Vector Spaces by 3Blue1Brown. This video offers an intuitive, visual introduction to vector spaces and subspaces.

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/k7RM-ot2NWy>



- MIT OpenCourseWare: Linear Independence, Basis, and Dimension. A lecture by Gilbert Strang, covering the key concepts of linear independence and basis. <https://ocw.mit.edu/courses/mathematics/18-06-linear-algebra-spring-2010/video-lectures/lecture-3/>

3 Python Programming Tasks

3.1 Practical Application: Introduction to NumPy

Remark. *NumPy is a fundamental library in Python for numerical computations, providing support for large multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on them.*

3.1.1 Basic Operations with Vectors and Matrices in NumPy

1. Creating Vectors and Matrices

```
import numpy as np
# Create a vector (1D array)
vector = np.array([1, 2, 3])
print(vector)
# Create a matrix (2D array)
matrix = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
print(matrix)
```

2. Vector Addition and Scalar Multiplication:

```
v1 = np.array([1, 2, 3])
v2 = np.array([4, 5, 6])
# Vector addition
sum_vectors = v1 + v2
print(sum_vectors) # Output: [5 7 9]
# Scalar multiplication
scalar_mult = 3 * v1
print(scalar_mult) # Output: [3 6 9]
```

3.2 Matrix Operations:

```
m1 = np.array([[1, 2], [3, 4]])
m2 = np.array([[5, 6], [7, 8]])
# Matrix addition
sum_matrices = m1 + m2
print(sum_matrices) # Output: [[ 6 8]
#
[10 12]]
```

```
# Matrix multiplication
product = np.dot(m1, m2)
print(product) # Output: [[19 22]
                # [43 50]]
```

3.3 Dot Product

```
# Dot product of two vectors
dot_product = np.dot(v1, v2)
print(dot_product) # Output: 32
```

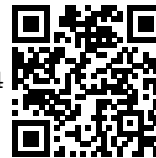
3.4 Norm of a Vector

```
# Compute the norm (length) of a vector
norm = np.linalg.norm(v1)
print(norm) # Output: 3.7416573867739413
```

3.5 Content Curated Video

- **Introduction to NumPy by DataCamp.** A tutorial on how to use NumPy for array operations, including basic vector and matrix manipulation.
<https://www.datacamp.com/community/tutorials/python-numpy-tutorial>
- Solving Real-World Problems with NumPy. A YouTube video demonstrating how NumPy can be used in practical, real-world scenarios.
Video Visit the URL below to view a video:

<https://www.youtube.com/embed/GB9ByFAIAH4>



3.6 Key Concepts:

- Vectors and vector spaces.
- Matrix operations (multiplication, inversion).
- Linear transformations in 2D/3D.

Tools: Python (NumPy, Matplotlib), Google Colab